

AFRICA CLINICAL RESEARCH NETWORK (ACRN)

Independent Outcomes Adjudication Committee (OAC)

SharePoint Adjudication Portal — Setup, Folder Architecture & Access-Control Specification

Studies in scope: PROTECT-Africa (EOPE, Protocol A202501 v1.2) and LOPE-Nigeria (LOPE, Protocol ACRN-202503 v1.1)
Companion to the Independent OAC Charter v2.0 and the OAC / Interim-Derivation Readiness WBD/WBS.

Document title	SharePoint Adjudication Portal Setup & Access-Control Specification — ACRN EOPE & LOPE
Version	1.0 (Draft for configuration & validation)
Effective date	Upon approval by Clinical Operations, QA and IT/SharePoint governance
Document owner	ACRN Clinical Operations (VP Clinical Operations / Project Management)
Applies to	Adjudication portal environment supporting blinded OAC endpoint adjudication
Governing documents	Independent OAC Charter v2.0; applicable ACRN Clin Ops & QA SOPs
Applicable standards	ICH E6(R2) GCP; EU Annex 11; US 21 CFR Part 11; GAMP 5; applicable national data-protection law
Classification	Confidential — Sponsor and study governance use
Related WBS refs	PRT-01 ... PRT-13 (Portal Configuration Tracker); eTMF zone 04.05 / 05

This is a controlled configuration document. Uncontrolled when printed.

1. Purpose, Scope and Design Objective

This specification defines how the ACRN Independent Outcomes Adjudication Committee (OAC) adjudication portal is provisioned on Microsoft SharePoint Online: the site architecture, document-library and folder structure, security groups, permission levels, access-control rules, and the technical controls that preserve blinding, study separation, attributability and inspection-readiness required by the OAC Charter v2.0.

The portal supports blinded, independent endpoint adjudication for two studies — PROTECT-Africa (EOPE) and LOPE-Nigeria (LOPE) — under a single OAC. It is a controlled-access clinical governance system, not a general collaboration site. Every design choice below is subordinate to one rule: adjudicators must never be able to reach a biomarker or point-of-care (POC) result, and the two studies must never be intermingled (Charter §1, §5.2, §7.3).

Out of scope

- The EDC, biomarker/POC laboratory systems, SDTM/ADaM statistical environment and the safety/SAE reporting pathway. These remain in their own validated systems; the portal receives only blinded, de-identified case content.
- Clinical care decisions, safety causality and SAE reporting — explicitly not OAC functions (Charter §2).

2. Design Principles

1. **Blinding by architecture, not by policy.** Biomarker/POC data is never ingested into the portal. Adjudicator-facing libraries are physically separate site content from staging and QC libraries, so an unblinding value has no path to a reviewer even on misconfiguration.
2. **Study separation.** EOPE and LOPE live in separate SharePoint sites with independent permission boundaries; cases, forms, logs and reports are never co-located (Charter §1, §7.3).
3. **Least privilege and need-to-know.** Access is granted through role-based security groups against libraries with unique (broken-inheritance) permissions — never through per-person ad-hoc grants.
4. **Independence.** Site investigators and study statisticians have no access to in-flight adjudication. QC-passed output is transferred out to the study database only after Chair sign-off (Charter §5.1, §10.3).
5. **Attributability and immutability.** Versioning is on everywhere; the Microsoft Purview unified audit log is enabled; deletion is blocked by retention labels. Every decision is date-stamped, attributable and reproducible (Charter §8, §11).
6. **Controlled external access.** Core members are external clinicians in Zimbabwe, Tanzania, Rwanda and Nigeria. They are onboarded as governed Entra ID (B2B) guests under Conditional Access, MFA and session controls — not via open external sharing.
7. **Validated and inspection-ready.** The portal is configured under change control and released only after IQ/OQ/UAT, consistent with Annex 11, 21 CFR Part 11 and GAMP 5.

3. Portal Architecture — Hub and Spoke

A single SharePoint hub associates four purpose-built sites. Separate sites (rather than one site with many libraries) are recommended because they give clean, auditable permission boundaries, contain study separation at the site level, and let data-loss-prevention, sensitivity-label and retention scoping differ per zone. The hub provides shared navigation, search scoping and a common theme; it does not grant access — each spoke controls its own membership.

Site (spoke)	Purpose	Primary access
OAC Governance & Administration	Charter, adjudication manual, membership/COI, training, agendas & minutes, SOPs, validation records.	Coordinator, Chair, Clin Ops, QA
EOPE Adjudication (PROTECT-Africa)	Blinded EOPE case packages, adjudication forms, queries, third-review, expedited/sentinel, adjudication log.	OAC voting members (blinded), Coordinator
LOPE Adjudication (LOPE-Nigeria)	Blinded LOPE case packages incl. normotensive reference cohort; forms, queries, log — fully separate from EOPE.	OAC voting members (blinded), Coordinator
OAC Restricted / Controlled	Chair e-signature & final sign-off, QC-passed output staged for transfer, unblinding-incident quarantine, eTMF export.	Chair, Coordinator, QA (tightly scoped)

Ingestion boundary: case content is assembled and blinded outside the portal and uploaded only by the Adjudication Coordinator into a staging library. No system integration writes biomarker/POC values into any portal site.

4. Security Groups and Role Mapping

Access is assigned exclusively through the SharePoint security groups below, each mapped to a Charter role. Members are added to groups only after the provisioning gate in Section 11 is satisfied. Non-voting roles cannot classify cases; the Coordinator and SMEs never adjudicate.

Security group	Charter role	Notes / constraints
OAC-Portal-Admin	IT / SharePoint governance	Full Control on configuration only. No access to case content. Change-controlled.
OAC-ClinOps-Admin	Clinical Operations	Manages governance site; approves access; does not adjudicate.
OAC-Coordinator	Adjudication Coordinator (non-voting)	Prepares/uploads blinded packages, runs pre-release check, manages logs & queries. No vote, no classification.
OAC-Chair	Chair & Chair-designate	Adjudicates, arbitrates deadlocks, final sign-off. Designate inherits when acting.
OAC-Voting-Members	Four core voting members	Blinded adjudication only. Per-case recusal/COI enforced via assignment (Section 10).
OAC-SME	Subject-matter experts (ad hoc, non-voting)	Time-boxed, case-scoped read access. Advisory only; no vote; no quorum.
OAC-QA-eTMF	Quality Assurance / TMF	Read + QC on governance, pre-release QC and restricted zones; blinding-integrity oversight.
OAC-Biostatistics	Study / independent statistician	No access to in-flight adjudication. Read only to 'Output for Transfer' post Chair sign-off.
OAC-Sponsor-Read	Sponsor representative	Informed. Read only to designated governance documents (charter, minutes).

5. Custom Permission Levels

SharePoint's default levels are replaced by purpose-built levels so that blinding and immutability are enforced by the permission model itself. Note in particular that no case-content group is granted Delete.

Level	Based on	Rights (key points)
Portal-Admin	Full Control	Full configuration control; excluded from case libraries by design.
Coordinator-Manage	Edit minus Delete	Add/edit/version items and set metadata; manage views; cannot hard-delete (retention-locked).
Adjudicate	Contribute (custom)	Add own adjudication form; view own submissions; edit own until locked; cannot see other reviewers' entries before concordance; no download/print where Restricted-View applies.
Restricted-View	Restricted View	View in-browser only — no download, print, copy or sync. Applied to blinded case packages and source documents.
QA-Review	Read + content approval	Read all in scope; run pre-release/QC approvals; annotate; no edit of adjudication decisions.
Read-Only	Read	View/read designated items only.
No-Access	—	Explicit denial; used to fence statisticians/sites from in-flight adjudication.

6. Folder and Library Structure

Each site uses document libraries with unique permissions (inheritance broken). Within libraries, folders provide navigation while access is governed at the library level and, for sensitive cases, at the item level. Case state is driven primarily by a managed-metadata Status column and views rather than by moving files, except where a permission boundary requires a physically separate library (staging vs published).

6.1 OAC Governance & Administration site

ACRN-OAC-Governance/
 └─ 00_Portal_Admin_Config/ (Portal-Admin only)

└─ 01_Charter_and_Adjudication_Manual/	└─ Charter_v2.0/	└─ Manual_current+superseded/
└─ 02_Membership_COI_Training/	└─ CVs/	└─ GCP_Certs/
└─ Training_Attestations/	└─ COI_Forms/	└─ Confidentiality/
└─ 03_Agendas_and_Minutes/	└─ EOPE_block/	└─ LOPE_block/
└─ 04_SOPs_Work_Instructions/		
└─ 05_Validation_IQ_OQ_UAT/	└─ Config_Spec/	└─ IQ/
	└─ OQ/	└─ UAT/
		└─ Access_Reviews/

6.2 EOPE Adjudication site (PROTECT-Africa)

Prefix **E1x** denotes EOPE. The two-library publish pattern (Staging → Pre-release QC → Ready-for-Review) is the core blinding gate.

ACRN-OAC-EOPE/

└─ E10_Case_Staging/	(Coordinator only – assembly & upload)
└─ E11_PreRelease_QC/	(Coordinator + QA – documented unblinding check)
└─ E12_Blinded_Case_Packages/	(Voting members – Restricted-View) [Ready for Review]
└─ folders by window: W1-2wk / W2-4wk	
└─ E13_Adjudication_Forms/	(Voting members – Adjudicate; reviewer-scoped)
└─ E14_Queries/	(Coordinator manage; members read own)
└─ E15_Discordant_ThirdReview/	(Coordinator assigns 3rd reviewer)
└─ E16_Adjudication_Log/	(Coordinator-Manage; QA read) – attributable, versioned
└─ E17_Expedited_Sentinel/	(out-of-cycle: maternal death, eclampsia, HELLP, stroke, DIC...)

6.3 LOPE Adjudication site (LOPE-Nigeria)

Structure mirrors EOPE with prefix **L1x**, plus a dedicated normotensive-reference-cohort area whose outcome-free status is confirmed by adjudication (Charter §3.2).

ACRN-OAC-LOPE/

└─ L10_Case_Staging/	└─ L11_PreRelease_QC/	└─ L12_Blinded_Case_Packages/
└─ L13_Adjudication_Forms/	└─ L14_Queries/	└─ L15_Discordant_ThirdReview/
└─ L16_Adjudication_Log/	└─ L17_Expedited_Sentinel/	
└─ L18_Normotensive_Reference_Cohort/	(reference-range confirmation, blinded)	

6.4 OAC Restricted / Controlled site

ACRN-OAC-Restricted/

└─ R20_Chair_SignOff_eSignature/	(Chair + Coordinator; validated e-signature)
└─ R21_Output_for_Transfer/	(post sign-off; Biostatistics read-only)
└─ R22_Unblinding_Incident_Quarantine/	(break-glass; QA + Clin Ops; access logged)
└─ R23_Archive_eTMF_Export/	(QC-passed, locked records for eTMF filing)

7. Blinding and Confidentiality Controls in SharePoint

These controls implement Charter §5.2 and §6 inside the platform. They are layered so no single failure exposes an unblinding value.

- **Source control at ingestion.** Only the Coordinator uploads, and only into E10/L10 Staging. Packages are assembled and de-identified (study IDs only) before upload; no biomarker/POC field, category or derived risk value is present.
- **Two-stage publish gate.** A package moves Staging → Pre-release QC, where the Coordinator runs the documented pre-release unblinding check and QA counter-checks. Only when 'Blinding Check Passed = Yes' is set does the package publish to the members' Blinded Case Packages library. Members have no access to Staging or QC libraries.
- **Restricted-View on case content.** Blinded packages and source documents are view-in-browser only — no download, print, copy or OneDrive sync — reducing the chance of content leaving the controlled boundary.
- **Sensitivity labels + IRM.** A 'Confidential — OAC' Microsoft Purview label applies encryption and do-not-forward; libraries are IRM-protected so downloaded copies (where permitted) remain rights-managed.
- **Data-loss-prevention (DLP).** A DLP policy scans for biomarker terms (e.g., sFlt-1, PlGF, sEng) and POC product names (e.g., CercaTest, GRAVIDAS, PEscreens) in items published to member libraries; a match blocks publication and raises an incident to QA/Clin Ops.
- **Reviewer isolation.** The Adjudicate level scopes each reviewer to their own form entries, so the two independent reviews cannot influence one another before concordance (Charter §6, §8).
- **Unblinding-incident handling.** Any suspected/actual unblinding is logged, moved to R22 quarantine under break-glass access, impact-assessed, reported to Clin Ops, and affected cases are re-adjudicated by unaffected members (Charter §5.2).

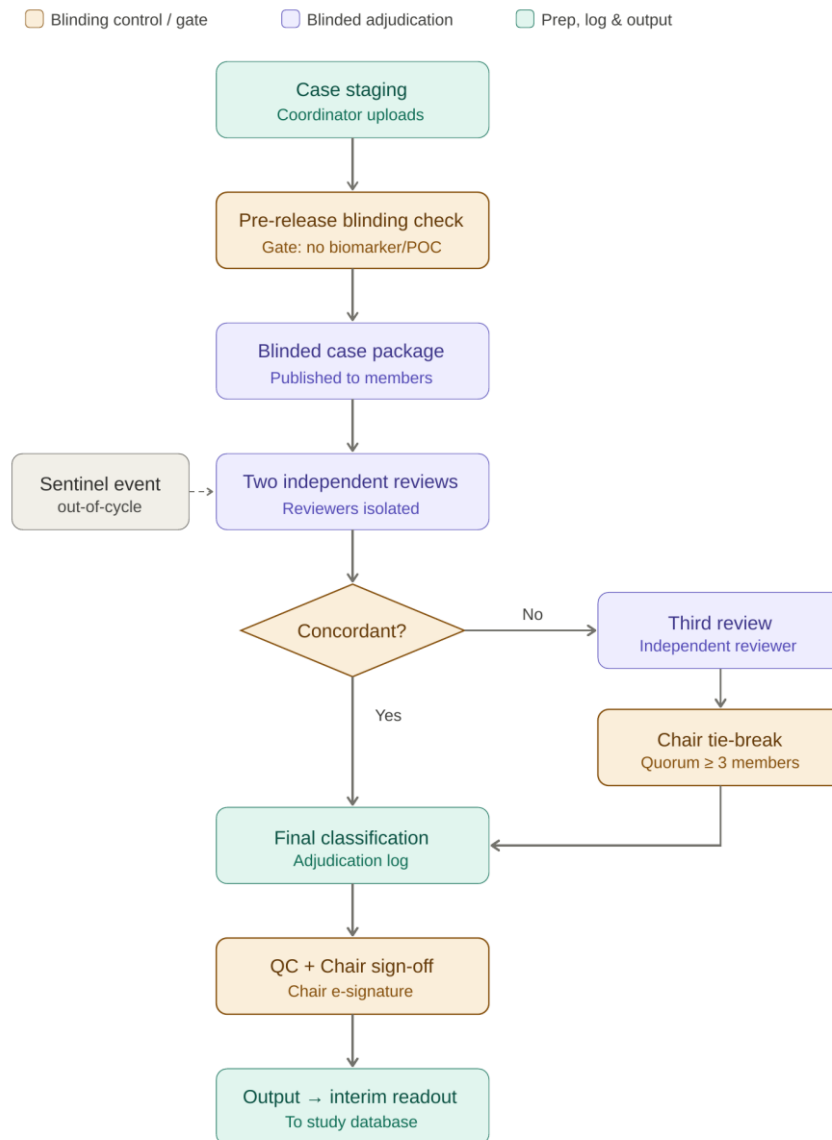
8. Case Flow Mapped to Libraries and States

The standardized case flow (Charter §6) is realised as metadata states and library transitions with an unbroken audit trail:

Step	Charter action	Library	Status metadata
1	Package preparation + pre-release check	E10/L10 Staging → E11/L11 QC	Draft → Pre-release QC
2	Publish blinded package	E12/L12 Blinded Case Packages	Ready for Review
3	Two independent blinded reviews	E13/L13 Adjudication Forms	Under Review (R1/R2)
4	Concordance → final	E16/L16 Adjudication Log	Concordant — Final
5	Discordance → third review	E15/L15 Third Review	Discordant — 3rd Review
6	Panel resolution / Chair tie-break	E16/L16 Log	Panel / Chair Tie-break
7	Query held / missing data	E14/L14 Queries	Query Open / Not Adjudicable
8	QC + Chair sign-off → transfer	R20 → R21 Output for Transfer	Signed-off — Released

8.1 Figure 1 — Blinded adjudication workflow

The schematic below traces a case end-to-end through the portal: blinded intake and the pre-release gate, dual independent blinded review, concordance or third-review/Chair tie-break resolution, then QC, Chair sign-off and transfer to the study database. Sentinel events enter out-of-cycle (Charter §6, §7.4, §8).



Amber = blinding control / decision gate **Purple** = blinded adjudication **Teal** = preparation, log & output

9. Content Types and Metadata

Study-specific content types are published from a Content Type Hub so EOPE and LOPE stay separate yet consistent. Structured columns make adjudication attributable and support endpoint-to-dataset traceability (Charter §10, §11).

Case Package content type

- Study (EOPE / LOPE); Case ID (study participant ID); Site code; Gestational age at onset; Prediction window; Cohort (derivation / validation / reference).
- Blinding Check Passed (Yes/No); Package Version; Date Ready; Assigned Reviewer 1 / 2 / 3; Recusals; Status; Expedited (Yes/No).

Adjudication Form content type

- Linked Case ID; Reviewer; Classification (PE / severe PE / eclampsia / HELLP; timing; EOPE-LOPE; severity; maternal & neonatal composites; relationship to PE); Criteria applied; Rationale; Date-stamp.

Query and Decision content types

- Query: Case ID; raised-by; question; target turnaround; response; resolution date. Decision/dissent recorded in the adjudication log with attributable, versioned entries.

10. Per-Case Recusal and Conflict-of-Interest Enforcement

Charter §5.3 requires that a member not adjudicate a case from a site where they have a conflict. Because SharePoint cannot infer clinical conflicts, enforcement is assignment-driven with a technical backstop:

- The Coordinator assigns reviewers per case and does not assign a conflicted member; assignment is captured in the Case Package metadata and drives a reviewer-scoped view so a member sees only their assigned, non-conflicted cases.
- For sensitive or boundary cases, item-level permissions physically remove a conflicted member from that item.
- Recusals are recorded in the adjudication log; a recused member does not count toward quorum for that case (Charter §8).

11. Access Provisioning, Review and Deprovisioning

Access gate (all must be satisfied before a member is added to any group): COI assessment cleared, confidentiality agreement signed, GCP evidence on file, and charter/manual training completed — the same gates tracked in the OAC Membership & COI Tracker.

- Joiner: Clin Ops raises access request → verified against the gate → Portal-Admin adds the person to the correct security group only. No direct, per-item grants outside the recusal backstop.
- SME (ad hoc): time-boxed access with an expiration date and case scope; auto-removed on expiry.
- Mover / recusal: group membership adjusted immediately on role change or a declared conflict.
- Leaver: access revoked on study exit; guest account disabled; action logged.
- Periodic review: quarterly access recertification by Clin Ops + QA; evidence filed in 05_Validation/Access_Reviews and the eTMF.

12. External Member Access and Tenant Security

Core members are external to the ACRN tenant and located across four countries. They are onboarded as Entra ID (B2B) guests — generic ('anyone'/link) external sharing is disabled tenant-wide for these sites.

- **Conditional Access:** MFA enforced; legacy authentication blocked; sign-in-risk and device policies applied; sessions time-boxed.
- **Session controls (Defender for Cloud Apps):** on unmanaged devices, download/print/copy are blocked (browser-only), reinforcing Restricted-View.
- **Guest governance:** guest access reviews and access expiration; guests limited to their assigned sites only.
- **Data residency & privacy:** case content uses de-identified study IDs (Charter §11). Tenant data region, and whether Multi-Geo is required, are confirmed with Sponsor/IT against applicable national data-protection law (e.g., Nigeria NDPA and Zimbabwe/Rwanda/Tanzania regimes). See open items.

13. Versioning, Audit Trail and Records Retention

- Major + minor versioning enabled on every library; all major versions retained; content approval on logs and forms.
- Microsoft Purview unified audit log enabled: who accessed/classified what, when — supporting the attributable audit trail (Charter §11).
- Deletion prevented: no case-content group holds Delete, and a retention label ('OAC-Adjudication — Retain, no-delete/edit') locks records for the protocol-defined retention period (to be confirmed with Sponsor/QA).
- Superseded charter/manual versions retained in version history; re-adjudication decisions retain both the revised and superseded decision (Charter §10.3).

14. Electronic Signature and Part 11 / Annex 11

Native SharePoint approvals do not, by themselves, satisfy 21 CFR Part 11 signature-manifestation requirements. Chair final sign-off (Charter §10.3) therefore uses a validated e-signature service (e.g., Adobe Acrobat Sign or DocuSign, Part 11 configuration) integrated at R20:

- The signed record carries the signer identity, the meaning of the signature (e.g., 'Approved for release to study database'), and a secure date-time stamp, with its own tamper-evident audit trail.
- The executed PDF is stored in R20 and the released output in R21; only QC-passed, signed output is transferred (Charter §10.3, §11).
- The signature workflow, along with the portal configuration, is covered by the validation package (Section 15).

15. Validation and Go-Live

The portal is treated as a GxP computerised system (GAMP 5, category 4 configured product) and released under change control. This maps directly to the readiness WBS (PRT-11 access testing, PRT-12 UAT, PRT-13 go-live).

8. Configuration specification (this document) approved by Clin Ops, QA and IT/SharePoint governance.
9. IQ — verify the built configuration matches this spec (sites, libraries, groups, permission levels, labels, DLP, retention).
10. **OQ — execute test scripts including negative tests:** a voting member cannot open Staging/QC; cannot access the other study's site; cannot download a Restricted-View package; a DLP-flagged biomarker term blocks publication; a statistician cannot see in-flight adjudication.
11. UAT — Coordinator and Chair test the end-to-end flow (upload → publish → dual review → discordance → third review → Chair sign-off → export).
12. Go-live approval — documented sign-off; access provisioned only for gate-cleared members; validation evidence filed to the eTMF.

16. eTMF Filing and Inspection Readiness

The following are filed as controlled records: this configuration specification; the exported access-control matrix and group membership; permission-level definitions; validation (IQ/OQ/UAT) evidence; access-review records; DLP/label/retention configuration; and the unblinding-incident log. These support the Charter §11 documentation standards and the portal-validation line in the TMF/eTMF Filing Tracker (eTMF zone 04.05).

17. Assumptions and Open Items for Confirmation

- Microsoft 365 licensing supports sensitivity labels, DLP, Conditional Access and Defender for Cloud Apps session control (E5 / compliance + AAD P2 or equivalent).
- Records-retention period to be confirmed with Sponsor/QA and set on the retention label.
- Choice of validated e-signature service (Adobe Acrobat Sign vs DocuSign) and its Part 11 configuration.
- Members onboarded as B2B guests vs sponsor-issued accounts; device management expectations for external members.
- Tenant data region / Multi-Geo decision against applicable national data-protection requirements for ZW, TZ, RW and NG.
- Confirmation that biomarker/POC data has no ingestion path into the portal from any connected system.

Prepared as a companion to the OAC Charter v2.0 and the OAC / Interim-Derivation Readiness WBD/WBS. For configuration by IT/SharePoint governance under change control; not effective until approved and validated.

Appendix A. Consolidated Access-Control Matrix

Legend: **FC** Portal-Admin (config) **M** Coordinator-Manage **A** Adjudicate **RV** Restricted-View **QA** QA-Review **R** Read-Only — No access

Library / resource	Portal Admin	ClinOps Admin	Coord.	Chair	Voting Members	SME (ad hoc)	QA / eTMF	Biostat (indep.)	Sponsor
OAC Governance & Administration site									
00 Portal Admin / Config	FC	—	—	—	—	—	R	—	—
01 Charter & Adjudication Manual	FC	M	M	R	R	R	QA	R	R
02 Membership / COI / Training	FC	M	M	R	—	—	QA	—	—
03 Agendas & Minutes (EOPE/LOPE)	FC	M	M	M	R	—	QA	—	R
04 SOPs / Work Instructions	FC	M	R	R	R	R	QA	R	R
05 Validation IQ/OQ/UAT / Access Reviews	FC	M	R	R	—	—	QA	—	—
EOPE Adjudication site (PROTECT-Africa) — LOPE site (L10–L18) mirrors these rules, separate membership									
E10 Case Staging	FC	—	M	—	—	—	QA	—	—
E11 Pre-release QC	FC	—	M	R	—	—	QA	—	—
E12 Blinded Case Packages	FC	—	M	RV	RV	RV	QA	—	—
E13 Adjudication Forms	FC	—	M	A	A	—	QA	—	—
E14 Queries	FC	—	M	R	R	—	QA	—	—
E15 Discordant / Third Review	FC	—	M	A	A	—	QA	—	—
E16 Adjudication Log	FC	—	M	R	R	—	QA	—	—
E17 Expedited / Sentinel	FC	—	M	A	A	—	QA	—	—
OAC Restricted / Controlled site									
R20 Chair Sign-off / e-Signature	FC	—	M	A	—	—	QA	—	—
R21 Output for Transfer (post sign-off)	FC	—	M	R	—	—	QA	R	—
R22 Unblinding-Incident Quarantine	FC	R	M	R	—	—	QA	—	—
R23 Archive / eTMF Export	FC	R	R	R	—	—	QA	—	—

Note: LOPE (L10–L18) applies identical rules to EOPE but with its own site, groups and separate membership so cases are never intermingled (Charter §7.3). 'Voting Members' access to case packages is Restricted-View (browser-only, no download) and is further narrowed per case by recusal/COI assignment (Section 10). Biostatistics gains Read only to R21 and only after Chair sign-off.

Appendix B. Naming Conventions and Provisioning Notes

Naming conventions

Object	Pattern	Example
Site	ACRN-OAC-{Zone}	ACRN-OAC-EOPE
Library	{Prefix}-{nn}_{Name}	E12_Blinded_Case_Packages
Case package	{STUDY}-{SITE}-{SUBJID}-{WINDOW}-v{n}	EOPE-ZW01-000123-W2-v1
Adjudication form	{CASEID}-ADJ-R{reviewer#}	EOPE-ZW01-000123-ADJ-R1
Query	{CASEID}-Q{nn}	LOPE-NG03-000456-Q02
Dates	ISO 8601 (YYYY-MM-DD)	2026-07-28

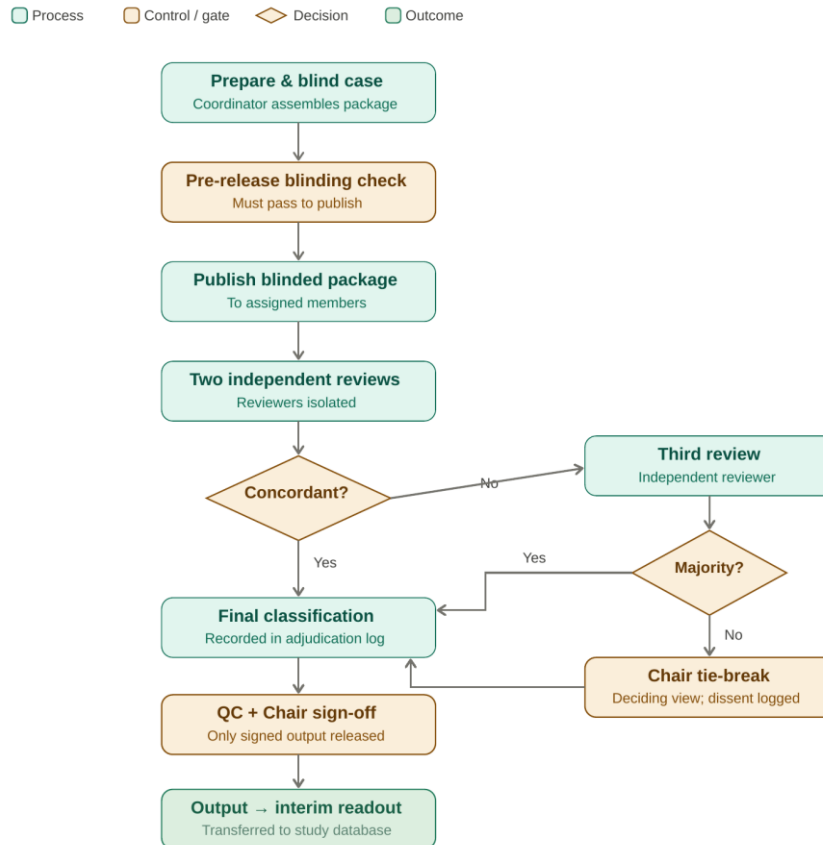
Provisioning approach

- Provision the four sites, libraries, content types, groups and permission levels via a repeatable PnP site template so the build is reproducible and re-verifiable at IQ.
- Break permission inheritance on every case-content library; assign only the security groups shown in Appendix A.
- Publish Case Package / Adjudication Form / Query content types from the Content Type Hub to keep EOPE and LOPE consistent yet separate.
- Apply the 'Confidential — OAC' sensitivity label, the OAC DLP policy and the 'OAC-Adjudication — Retain' retention label at site provisioning, before any content is uploaded.

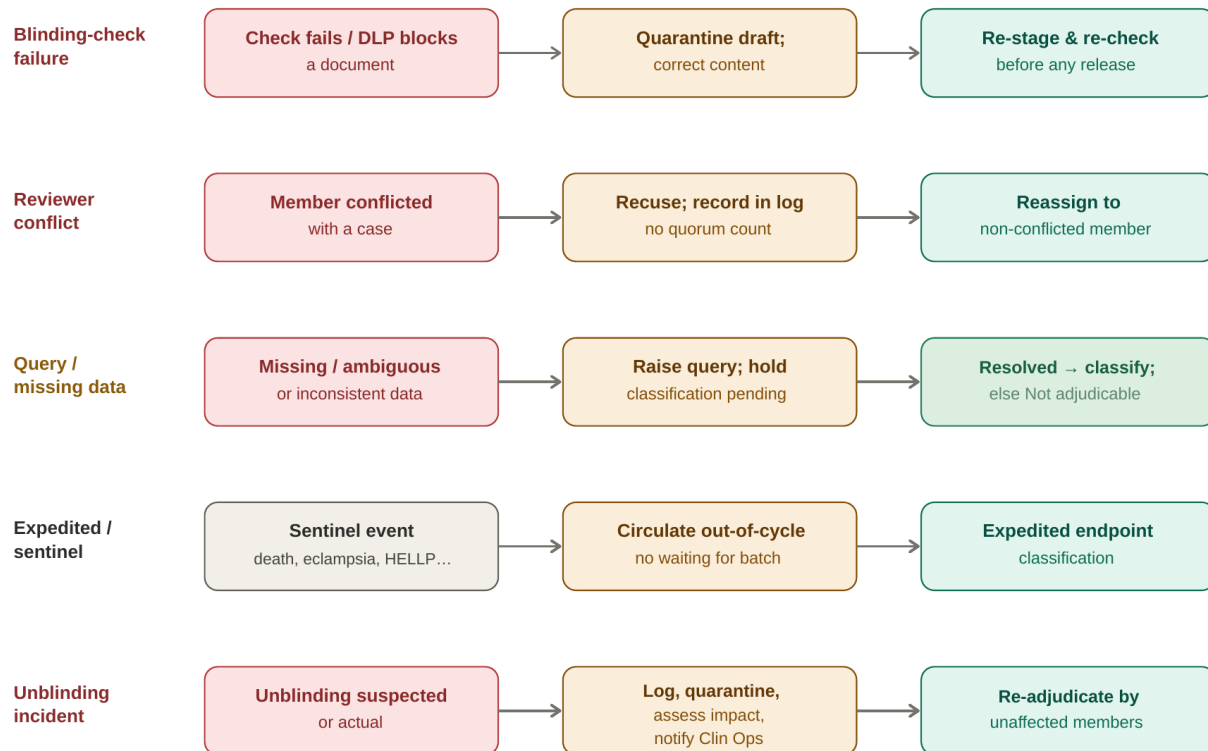
Appendix C. Adjudication Process — All Scenarios

Figure 2 shows the core decision flow every case follows; Figure 3 shows the exception and special-path scenarios. Together they cover every route a case can take. All exception paths rejoin the core flow and are captured in the attributable, versioned audit trail (Charter §6, §7.4, §8, §10.3).

Figure 2 — Core adjudication decision flow



Concordant → final. Discordant → third review; a majority resolves, otherwise the Chair's deciding view. Then QC, Chair sign-off and transfer.

Figure 3 — Exception and special-path scenarios

Expedited review is for endpoint classification only — it does not replace or delay SAE / safety reporting, which runs on its own clock.

All exception paths rejoin the core flow (Figure 2) and are captured in the versioned, attributable audit trail.

Blinding-check failure, reviewer conflict/recusal, query/missing data, expedited sentinel review, and unblinding incidents — each rejoins the core flow.